

Question	Answer	Marks
4(a)	(when two or more) waves meet/overlap (at a point)	B1
	(resultant) displacement is sum of the individual displacements	B1
4(b)(i)	Fringe width, $x = 3.2 \times 10^{-2} / 8$ $= 4.0 \times 10^{-3} \text{ (m)}$	C1
	$D = ax / \lambda$ $= (4.0 \times 10^{-3} \times 0.16 \times 10^{-3}) / 7.2 \times 10^{-7}$	C1
	$= 0.89 \text{ m}$	A1

Question	Answer	Marks
4(b)(ii)	Curved line with a negative gradient of decreasing magnitude throughout, from slit separation 0.04 mm to 0.16 mm	B1
	Line of negative gradient ending at (0.16, 0.4), from slit separation 0.04 mm	B1
	Line of negative gradient passing through (0.08, 0.8) <u>and</u> (0.04, 1.6)	B1

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